Lesson 12: Markov Chains

AI/ML Mathematics

Contents

1 Motivation and Overview **2**

2 References **2**

3 Markov Chains and Transition Matrices **2**

3.1 Chapman–Kolmogorov Equation 3

4 Classification of States **3**

5 Stationary Distributions **4**

6 Ergodic Theorem for Finite Markov Chains **5**

7 Detailed Balance and Reversibility **6**

8 Exercises **7**

9 Summary **7**

10 Further Reading **8**

1 Motivation and Overview

A Markov chain is a stochastic process whose future evolution depends on its past only through its current state. This "memoryless" property is the defining structural assumption of Markov Decision Processes (MDPs), and every fixed policy in an MDP induces a Markov chain on the state space. Understanding the long-run behavior of Markov chains—convergence to a stationary distribution, ergodic time averages, and the role of reversibility—is therefore essential for RL theory.

This lesson develops the core theory of discrete-time Markov chains on finite or countable state spaces: transition matrices and the Chapman–Kolmogorov equation; classification of states (communication, recurrence, periodicity); stationary distributions (existence, uniqueness, mean return times); the ergodic theorem; and detailed balance.

Why This Matters for RL

- **MDPs as controlled Markov chains:** Under a fixed policy π, the state sequence $\{S_t\}$ in an MDP is a Markov chain with transition matrix $P^\pi(s'|s) = \sum_a \pi(a|s)P(s'|s,a)$.
- **Policy evaluation:** The convergence of TD(0) and linear TD relies on the induced Markov chain being ergodic, so that time averages along a single trajectory converge to expectations under the stationary distribution d^π.
- **MCMC in model-based RL:** Markov Chain Monte Carlo (MCMC) methods, used for posterior sampling in Bayesian RL, construct reversible Markov chains whose stationary distribution is the target posterior. Detailed balance is the key design condition.

Prerequisites. Lesson 5 (expectation, conditional probability), Lesson 7 (independence), Lesson 10 (modes of convergence), Lesson 11 (SLLN, dominated convergence).

2 References

The following sources complement this lesson:

- Norris [1997], Chapters 1–3: a self-contained treatment of discrete-time Markov chains covering all topics in this lesson.
- Levin and Peres [2017], Chapters 1–4: Markov chains and mixing times with emphasis on convergence rates and coupling arguments.
- Durrett [2019], Section 6.4: Markov chains in the context of probability theory.
- Grimmett and Stirzaker [2001], Chapter 6: Markov chains with many worked examples.

3 Markov Chains and Transition Matrices

Definition 3.1 (Markov Chain)**.** A sequence of random variables $\{X_t\}_{t\geq 0}$ taking values in a finite or countable state space $\mathcal{S}$ is a *(discrete-time) Markov chain* if for all $t \geq 0$ and all states $s_0, \ldots, s_t, s_{t+1} \in \mathcal{S}$:

$$\mathbb{P}(X_{t+1} = s_{t+1} \mid X_t = s_t, \ldots, X_0 = s_0) = \mathbb{P}(X_{t+1} = s_{t+1} \mid X_t = s_t). \tag{1}$$

The right-hand side is called the *transition probability* at time t.

Definition 3.2 (Time-Homogeneous Transition Matrix)**.** A Markov chain is *time-homogeneous* if $\mathbb{P}(X_{t+1} = s' \mid X_t = s) = P(s'|s)$ does not depend on t. The *transition matrix* $\mathbf{P} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{S}|}$ has entries $P_{ss'} = P(s'|s)$, with $P_{ss'} \geq 0$ and $\sum_{s'} P_{ss'} = 1$ for all s. Such a matrix is called *(row-)stochastic*. Throughout, all Markov chains are time-homogeneous unless stated otherwise.

Example 3.3 (Two-State Markov Chain)**.** Let $\mathcal{S} = \{0, 1\}$ with transition matrix

$$\mathbf{P} = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}, \quad \alpha, \beta \in (0, 1). \tag{2}$$

State 0 transitions to 1 with probability α, and state 1 transitions to 0 with probability β. This chain is irreducible (both states communicate) and aperiodic (self-loops exist since $\alpha, \beta < 1$).

3.1 Chapman–Kolmogorov Equation

Chapman–Kolmogorov Equation

Theorem 3.4 (Chapman–Kolmogorov Equation)**.** *For a time-homogeneous Markov chain with transition matrix* $\mathbf{P}$*, the* k*-step transition probability* $P^{(k)}(s, s') = \mathbb{P}(X_{t+k} = s' \mid X_t = s)$ *satisfies*

$$P^{(n+m)}(s, s') = \sum_{u \in \mathcal{S}} P^{(n)}(s, u)\, P^{(m)}(u, s'). \tag{3}$$

In matrix notation: $\mathbf{P}^{n+m} = \mathbf{P}^n \mathbf{P}^m$.

Proof. By the law of total probability and the Markov property:

$$\begin{aligned} P^{(n+m)}(s, s') &= \mathbb{P}(X_{n+m} = s' \mid X_0 = s) \\ &= \sum_u \mathbb{P}(X_{n+m} = s' \mid X_n = u)\, \mathbb{P}(X_n = u \mid X_0 = s) \\ &= \sum_u P^{(m)}(u, s')\, P^{(n)}(s, u), \end{aligned} \tag{4}$$

which is the (s, s') entry of $\mathbf{P}^n \mathbf{P}^m$. The second equality uses the Markov property (1). □

4 Classification of States

Definition 4.1 (Accessibility and Communication)**.** State s' is *accessible* from state s (written $s \to s'$) if $P^{(n)}(s, s') > 0$ for some $n \geq 0$. States s and s' *communicate* (written $s \leftrightarrow s'$) if $s \to s'$ and $s' \to s$.

Proposition 4.2 (Communication is an Equivalence Relation)**.** *The relation* $\leftrightarrow$ *is an equivalence relation on* $\mathcal{S}$*, partitioning the state space into* communicating classes.

Proof. Reflexive: $P^{(0)}(s, s) = 1 > 0$, so $s \to s$ and hence $s \leftrightarrow s$.

Symmetric: By definition, $s \leftrightarrow s'$ means $s \to s'$ and $s' \to s$, which is symmetric.

Transitive: Suppose $s \leftrightarrow u$ and $u \leftrightarrow s'$. Then there exist n_1, n_2 with $P^{(n_1)}(s, u) > 0$ and $P^{(n_2)}(u, s') > 0$. By Chapman–Kolmogorov:

$$P^{(n_1+n_2)}(s, s') \geq P^{(n_1)}(s, u)\, P^{(n_2)}(u, s') > 0, \tag{5}$$

so $s \to s'$. Similarly, $s' \to s$, and therefore $s \leftrightarrow s'$. □

Definition 4.3 (Irreducible Chain)**.** A Markov chain is *irreducible* if there is a single communicating class, i.e., $s \leftrightarrow s'$ for all $s, s' \in \mathcal{S}$.

Definition 4.4 (Recurrence and Transience)**.** Let $\tau_s = \inf\{t \geq 1 : X_t = s\}$ be the *first return time* to state s (with $\inf \emptyset = \infty$). State s is:

- *Recurrent* if $\mathbb{P}(\tau_s < \infty \mid X_0 = s) = 1$.
- *Transient* if $\mathbb{P}(\tau_s < \infty \mid X_0 = s) < 1$.

Theorem 4.5 (Recurrence Criterion)**.** *State s is recurrent if and only if $\sum_{n=0}^{\infty} P^{(n)}(s,s) = \infty$.*

Proof. Let $N_s = \sum_{t=1}^{\infty} \mathbf{1}_{\{X_t=s\}}$ be the total number of returns to s starting from $X_0 = s$. Then

$$\mathbb{E}[N_s \mid X_0 = s] = \sum_{t=1}^{\infty} \mathbb{P}(X_t = s \mid X_0 = s) = \sum_{t=1}^{\infty} P^{(t)}(s,s). \tag{6}$$

Case 1: s is transient. Let $p = \mathbb{P}(\tau_s < \infty \mid X_0 = s) < 1$. By the strong Markov property, each return to s is an independent "fresh start," so the probability of returning at least k times is p^k. Thus N_s has a geometric distribution: $\mathbb{P}(N_s \geq k \mid X_0 = s) = p^k$, giving $\mathbb{E}[N_s \mid X_0 = s] = p/(1-p) < \infty$. By (6), $\sum_{t=1}^{\infty} P^{(t)}(s,s) < \infty$.

Case 2: s is recurrent. If $p = 1$, then by the same argument, $\mathbb{P}(N_s \geq k \mid X_0 = s) = 1$ for every k, so $N_s = \infty$ a.s. and $\mathbb{E}[N_s \mid X_0 = s] = \infty$. By (6), $\sum_{t=1}^{\infty} P^{(t)}(s,s) = \infty$.

□

Proposition 4.6 (Recurrence is a Class Property)**.** *If s is recurrent and $s \leftrightarrow s'$, then s' is recurrent.*

Proof. Since $s \leftrightarrow s'$, there exist integers $m, n \geq 0$ with $P^{(m)}(s',s) > 0$ and $P^{(n)}(s,s') > 0$. For any $k \geq 0$, Chapman–Kolmogorov gives

$$P^{(m+k+n)}(s',s') \geq P^{(m)}(s',s)\, P^{(k)}(s,s)\, P^{(n)}(s,s'). \tag{7}$$

Let $c = P^{(m)}(s',s) \cdot P^{(n)}(s,s') > 0$. Summing over k:

$$\sum_{k=0}^{\infty} P^{(m+k+n)}(s',s') \geq c \sum_{k=0}^{\infty} P^{(k)}(s,s) = \infty, \tag{8}$$

since s is recurrent. Therefore $\sum_{t=0}^{\infty} P^{(t)}(s',s') = \infty$, and s' is recurrent by Theorem 4.5. □

5 Stationary Distributions

Definition 5.1 (Stationary Distribution)**.** A probability distribution $\boldsymbol{\pi} = (\pi(s))_{s\in\mathcal{S}}$ is *stationary* (or *invariant*) for a Markov chain with transition matrix $\mathbf{P}$ if

$$\pi(s') = \sum_{s\in\mathcal{S}} \pi(s)\, P(s'|s) \quad \text{for all } s' \in \mathcal{S}. \tag{9}$$

In vector notation: $\boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top \mathbf{P}$, i.e., $\boldsymbol{\pi}$ is a left eigenvector of $\mathbf{P}$ with eigenvalue 1.

Definition 5.2 (Aperiodic Chain)**.** The *period* of state s is $d(s) = \gcd\{n \geq 1 : P^{(n)}(s,s) > 0\}$. State s is *aperiodic* if $d(s) = 1$. An irreducible chain is aperiodic if any (equivalently, all) of its states are aperiodic.

Proposition 5.3 (Period is a Class Property)**.** *If $s \leftrightarrow s'$, then $d(s) = d(s')$.*

Proof. Let $s \to s'$ in m steps and $s' \to s$ in n steps. Then $P^{(m+n)}(s,s) \geq P^{(m)}(s,s')P^{(n)}(s',s) > 0$, so $d(s)$ divides $m+n$.

For any k with $P^{(k)}(s',s') > 0$, we have $P^{(m+k+n)}(s,s) \geq P^{(m)}(s,s')P^{(k)}(s',s')P^{(n)}(s',s) > 0$, so $d(s)$ divides $m+k+n$. Since $d(s)$ divides both $m+n$ and $m+k+n$, it divides k. This holds for all such k, so $d(s)$ divides $d(s')$. By symmetry, $d(s')$ divides $d(s)$, hence $d(s) = d(s')$. □

Definition 5.4 (Positive Recurrence)**.** A recurrent state s is *positive recurrent* if $\mathbb{E}[\tau_s \mid X_0 = s] < \infty$ (finite expected return time). It is *null recurrent* if it is recurrent but $\mathbb{E}[\tau_s \mid X_0 = s] = \infty$.

Stationary Distribution Theorem

Theorem 5.5 (Existence and Uniqueness of Stationary Distributions)**.** *For a finite, irreducible Markov chain:*

1. *The chain is positive recurrent.*
2. *There exists a unique stationary distribution* $\boldsymbol{\pi}$ *with* $\pi(s) > 0$ *for all* s*.*
3. $\pi(s) = 1/\mathbb{E}[\tau_s \mid X_0 = s]$ *for each* $s \in \mathcal{S}$*.*

Proof. **Part (1):** In a finite irreducible chain, all states communicate. At least one must be recurrent (if all were transient, $\sum_t P^{(t)}(s,s) < \infty$ for each s, contradicting $\sum_s P^{(t)}(s_0,s) = 1$ summed over t). By Proposition 4.6, all states are recurrent. Finiteness of the state space then gives positive recurrence.

Part (2), existence: Fix $s_0 \in \mathcal{S}$ and define $\mu(s') = \mathbb{E}[\sum_{t=0}^{\tau_{s_0}-1} \mathbf{1}_{\{X_t = s'\}} \mid X_0 = s_0]$, the expected visits to s' during one cycle from s_0. Conditioning on the step after each visit shows that the normalized vector $\pi(s) = \mu(s)/\sum_u \mu(u)$ satisfies $\boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top \mathbf{P}$. Since $\sum_u \mu(u) = \mathbb{E}[\tau_{s_0} \mid X_0 = s_0] < \infty$, $\boldsymbol{\pi}$ is a well-defined probability distribution.

Part (2), uniqueness: If $\boldsymbol{\pi}'$ is another stationary distribution, then $\pi'(s') = \sum_s \pi'(s) P^{(n)}(s,s')$ for all n. Taking Cesaro averages and using $\frac{1}{N}\sum_{n=0}^{N-1} P^{(n)}(s,s') \to \pi(s')$ (from renewal theory) gives $\pi'(s') = \pi(s')$.

Part (3): By symmetry (choosing s as the reference state), $\pi(s) = 1/\mathbb{E}[\tau_s \mid X_0 = s]$. □

Stationary Distribution in Policy Evaluation

Under a fixed policy π in a finite MDP, the induced chain has transition matrix $P^\pi(s'|s) = \sum_a \pi(a|s) P(s'|s,a)$. If irreducible, its unique stationary distribution d^π (with $d^\pi(s) > 0$) weights the Mean Squared Value Error $\overline{\mathrm{VE}}(\mathbf{w}) = \sum_s d^\pi(s)(V^\pi(s) - \hat{V}(s;\mathbf{w}))^2$ used in linear TD convergence proofs.

6 Ergodic Theorem for Finite Markov Chains

Definition 6.1 (Ergodic Chain)**.** A finite Markov chain is *ergodic* if it is irreducible and aperiodic.

Ergodic Theorem

Theorem 6.2 (Ergodic Theorem for Finite Markov Chains)**.** *If* $\{X_t\}$ *is an ergodic finite Markov chain with stationary distribution* $\boldsymbol{\pi}$*, then:*

1. ***Distribution convergence:*** *For any initial distribution,* $\mathbb{P}(X_t = s) \to \pi(s)$ *as* $t \to \infty$*, for every* $s \in \mathcal{S}$*.*
2. ***Time-average convergence:*** *For any function* $f : \mathcal{S} \to \mathbb{R}$*:*

$$\frac{1}{T}\sum_{t=0}^{T-1} f(X_t) \xrightarrow{a.s.} \sum_{s\in\mathcal{S}} \pi(s)\, f(s) = \mathbb{E}_\pi[f]. \tag{10}$$

Proof of Part (1). Aperiodicity and irreducibility imply there exists $n_0 \geq 1$ with $P^{(n)}(s,s') > 0$ for all s, s' and $n \geq n_0$. Let $\delta = \min_{s,s'} P^{(n_0)}(s,s') > 0$ and $N = |\mathcal{S}|$. Write the n_0-step transition as $P^{(n_0)}(s,\cdot) = N\delta \cdot \boldsymbol{q}(\cdot) + (1 - N\delta)\cdot \boldsymbol{r}_s(\cdot)$, where $\boldsymbol{q}$ is a common distribution. This gives a contraction in total variation with $\rho = 1 - N\delta < 1$:

$$\|\boldsymbol{\mu}^\top \mathbf{P}^{n_0} - \boldsymbol{\nu}^\top \mathbf{P}^{n_0}\|_{\mathrm{TV}} \leq \rho\, \|\boldsymbol{\mu} - \boldsymbol{\nu}\|_{\mathrm{TV}}. \tag{11}$$

Setting $\boldsymbol{\nu} = \boldsymbol{\pi}$ and iterating: $\|\boldsymbol{\mu}_0^\top \mathbf{P}^{kn_0} - \boldsymbol{\pi}^\top\|_{\mathrm{TV}} \le \rho^k \to 0$. Since total variation is non-increasing under $\mathbf{P}$, convergence holds for all t. □

Proof of Part (2). Let $N_T(s) = \sum_{t=0}^{T-1} \mathbf{1}_{\{X_t=s\}}$. The successive return times to s are i.i.d. with mean $1/\pi(s)$, so the SLLN (Lesson 11, Theorem 4.2) gives $N_T(s)/T \xrightarrow{\text{a.s.}} \pi(s)$. Since $\frac{1}{T}\sum_t f(X_t) = \sum_s f(s)N_T(s)/T$ and $\mathcal{S}$ is finite, the result follows. □

Ergodic Theorem and On-Policy Learning

The ergodic theorem justifies on-policy learning from a single trajectory: expectations under d^π (appearing in the SA convergence analysis of TD) are approximated by time averages, which converge a.s. by Part (2).

7 Detailed Balance and Reversibility

Definition 7.1 (Detailed Balance / Reversibility)**.** A Markov chain with transition matrix $\mathbf{P}$ satisfies *detailed balance* with respect to a distribution $\boldsymbol{\pi}$ if

$$\pi(s)\,P(s'|s) = \pi(s')\,P(s|s') \quad \text{for all } s, s' \in \mathcal{S}. \tag{12}$$

A chain satisfying detailed balance is called *reversible*.

Detailed Balance Implies Stationarity

Proposition 7.2 (Detailed Balance Implies Stationarity)**.** *If $\boldsymbol{\pi}$ satisfies detailed balance (12) with respect to $\mathbf{P}$, and $\sum_s \pi(s) = 1$ with $\pi(s) \ge 0$, then $\boldsymbol{\pi}$ is a stationary distribution for $\mathbf{P}$.*

Proof. Sum both sides of (12) over s: $\sum_s \pi(s)P(s'|s) = \sum_s \pi(s')P(s|s') = \pi(s')\sum_s P(s|s') = \pi(s')$, where the last step uses $\sum_s P(s|s') = \sum_s P_{s's} = 1$ (row sums of $\mathbf{P}$). This is the stationarity equation (9). □

Remark 7.3 (Converse Fails). Stationarity does *not* imply detailed balance. The three-state cycle $1 \to 2 \to 3 \to 1$ has stationary distribution $\pi = (1/3, 1/3, 1/3)$, but $\pi(1)P(2|1) = 1/3 \neq 0 = \pi(2)P(1|2)$.

Example 7.4 (Metropolis–Hastings Chain)**.** The Metropolis–Hastings algorithm constructs a reversible chain with prescribed stationary distribution $\boldsymbol{\pi}$. Given proposal $Q(s'|s)$, the acceptance probability is $\alpha(s, s') = \min(1,\ \pi(s')Q(s|s')/[\pi(s)Q(s'|s)])$. The resulting transition $P(s'|s) = Q(s'|s)\alpha(s, s')$ (for $s' \neq s$) satisfies detailed balance: $\pi(s)Q(s'|s)\alpha(s, s') = \min(\pi(s)Q(s'|s),\ \pi(s')Q(s|s')) = \pi(s')Q(s|s')\alpha(s', s)$.

MCMC in Model-Based RL

In Bayesian model-based RL, posterior sampling (e.g., Thompson sampling) often uses MCMC methods such as Metropolis–Hastings. Detailed balance ensures the chain's stationary distribution equals the target posterior, and the ergodic theorem guarantees convergence of sample averages to posterior expectations.

8 Exercises

Exercise 8.1 ((*)). For the two-state Markov chain of Example 3.3 with parameters $\alpha, \beta \in (0,1)$:

(a) Find the stationary distribution $\boldsymbol{\pi}$.

(b) Verify that the chain satisfies detailed balance with respect to $\boldsymbol{\pi}$.

(c) Compute $P^{(n)}(0,0)$ by diagonalizing $\mathbf{P}$, and verify convergence to $\pi(0)$.

Exercise 8.2 ((*)). A Markov chain on $\mathcal{S} = \{1,2,3\}$ has transition matrix

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Identify the communicating classes, classify each state as recurrent or transient, and find all stationary distributions.

Exercise 8.3 ((**)). A Markov chain on $\mathcal{S} = \{0,1,2,3\}$ has transitions $0 \to 1 \to 2 \to 3 \to 0$, each with probability 1.

(a) Show the chain is irreducible with period $d = 4$.

(b) Does $P^{(n)}(0,0)$ converge as $n \to \infty$?

(c) Show that the Cesaro averages $\frac{1}{N}\sum_{n=0}^{N-1} P^{(n)}(0,s)$ converge to $1/4$ for each s.

Exercise 8.4 ((**)). **(Gambler's Ruin.)** A gambler starts with k dollars and plays a fair coin-toss game, winning or losing \$1 each round, until reaching \$0 (ruin) or \$$N$ (target). Model this as a Markov chain on $\mathcal{S} = \{0,1,\ldots,N\}$ with absorbing states 0 and N.

(a) Write down the transition matrix.

(b) Identify the communicating classes and classify each as recurrent or transient.

(c) Compute the probability of ruin starting from state k. *Hint:* Set up a recurrence for the ruin probability r_k using the Markov property.

Exercise 8.5 ((***)). For an ergodic finite Markov chain with N states and contraction factor ρ from Theorem 6.2(1), show that the mixing time satisfies $t_{\text{mix}}(\varepsilon) \le n_0\lceil \ln(1/\varepsilon)/\ln(1/\rho)\rceil$. Compute ρ and bound $t_{\text{mix}}(0.01)$ for the two-state chain with $\alpha = \beta = 0.1$.

Exercise 8.6 ((***)). A two-state MDP has $\mathcal{S} = \{A, B\}$, $\mathcal{A} = \{a_1, a_2\}$, with $P(B|A,a_1) = 0.3$, $P(A|B,a_1) = 0.6$, $P(B|A,a_2) = 0.7$, $P(A|B,a_2) = 0.2$. Policy π chooses a_1 in A and a_2 in B. (a) Write $\mathbf{P}^\pi$. (b) Find d^π. (c) Is the chain reversible?

9 Summary

- A **Markov chain** is a memoryless stochastic process: the future depends on the past only through the current state.
- The **transition matrix** $\mathbf{P}$ encodes one-step probabilities; k-step probabilities are given by $\mathbf{P}^k$ (Chapman–Kolmogorov).
- **Communication classes** partition the state space; a chain is **irreducible** if it has one class.
- A state is **recurrent** iff $\sum_n P^{(n)}(s,s) = \infty$; recurrence and transience are class properties.

- A finite irreducible chain has a unique **stationary distribution** $\boldsymbol{\pi}$ with $\pi(s) = 1/\mathbb{E}[\tau_s]$.
- An **ergodic** (irreducible + aperiodic) chain converges to $\boldsymbol{\pi}$ in distribution, and time averages converge a.s. to $\boldsymbol{\pi}$-expectations.
- **Detailed balance** $\pi(s)P(s'|s) = \pi(s')P(s|s')$ implies stationarity and is the design principle behind MCMC methods.

10 Further Reading

- **Lesson 11** provides the SLLN and DCT used in the proofs of the ergodic theorem and the well-definedness of expected return times.
- **Lesson 13** on martingale convergence provides tools for analyzing Markov chains via optional stopping and martingale transforms.
- **Lesson 14** on stochastic approximation uses the Markov chain theory developed here (especially the ergodic theorem) for the ODE-method convergence proof.
- Norris [1997]: the standard reference, covering finite and countable state spaces with complete proofs.
- Levin and Peres [2017]: convergence rates and mixing times via coupling and spectral methods.
- Grimmett and Stirzaker [2001], Chapter 6: concise introduction with many applications.

References

Rick Durrett. *Probability: Theory and Examples.* Cambridge University Press, 5th edition, 2019.

Geoffrey R. Grimmett and David R. Stirzaker. *Probability and Random Processes.* Oxford University Press, 3rd edition, 2001.

David A. Levin and Yuval Peres. *Markov Chains and Mixing Times.* American Mathematical Society, 2nd edition, 2017.

James R. Norris. *Markov Chains.* Cambridge University Press, 1997.